Task 1: Laplace mechanism and counting queries (4 points)

Output:

```
--- Task 1: Laplace Mechanism and Counting Query ---
True count (people > 29):   22851
DP count (ε=ln(2)):         22851.36
Noise added:                +0.36
Sensitivity:                1
Epsilon:                    0.6931
```

a) What is the sensitivity of this query, and why?

The sensitivity of this query is **1**.

Reasoning:

Sensitivity (Δf) is defined as the maximum amount the query output can change when a single record is added to or removed from the dataset.

Since this is a **counting query** ("count people over 29"), adding or removing a single individual can change the count by at most **1**.

- If the individual is over 29, the count changes by +1 or -1.
- If the individual is 29 or younger, the count changes by 0.

Therefore, the maximum possible change (global sensitivity) is **1**.

Task 2: Contingency tables (4 points)

Output:

```
--- Task 2: Contingency Tables ---
Generating DP table for Relationship vs Race (epsilon=0.3)...

DP Table Results (First 5 rows):
Race             Amer-Indian-Eskimo  Asian-Pac-Islander       Black      Other         White
Relationship
Husband                   95.167350          409.380670  672.881273  80.642324  11939.583741
Not-in-family             77.338661          213.033622  813.103729  70.973335   7123.418962
Other-relative            16.737867           82.615693  165.041127  26.097454    691.715908
Own-child                 49.129319          173.074441  552.022975  41.392332   4250.247648
Unmarried                 48.988095           94.179651  759.458602  33.113607   2495.150188
```

(a) Does parallel composition apply?

Answer: Yes.

Why?

The contingency table splits people into groups. Each group is one cell. One person goes into only ONE cell. The cells do not share people.

- **Example:**
  - Person A is "Husband" + "White" → goes to cell (Husband, White)
  - Person B is "Wife" + "Black" → goes to cell (Wife, Black)
  - They are in different cells.

When we add or remove one person, only **ONE** cell changes. The other cells stay the same.

This means:

- We can use the full **ε** for each cell.
- Total privacy cost = **ε** (NOT **ε** × number of cells).
- This is called "parallel composition."

(b) Does the number of variables matter?

For Privacy Cost: NO

The total privacy cost is always ϵ. This does not change.

- 2 variables → privacy cost = **ε**
- 3 variables → privacy cost = **ε**
- 4 variables → privacy cost = **ε**

Why? One person still changes only one cell, regardless of the table size. So the total cost stays **ε.**

For Accuracy: YES

More variables = worse accuracy.

Why?

- More variables means more cells (e.g., 2 variables = 25 cells; 3 variables = 125 cells).
- More cells mean fewer people in each cell.
- We add the **same amount of noise** to every cell (because ϵ is the same).

Example:

- **Large Cell:** 1000 people + noise of 3 → error is small (0.3%).
- **Small Cell:** 50 people + noise of 3 → error is large (6%).

Summary:

- **Privacy cost:** Stays the same (**ε**).
- **Accuracy:** Gets worse with more variables.

Task 3: Differentially private selections from sets (4 points)

Output:

```
TASK 3: Differentially Private Selection
Total records: 30718
Privacy budget (ε): 0.05
Number of unique occupations: 14

------------------------------------------------------------
RESULTS:
True most common occupation:  Prof-specialty

DP selected occupation:       Exec-managerial
DP noisy count:               4099.41
True count of selected:       4066

Correct selection: NO

------------------------------------------------------------
TOP 5 OCCUPATIONS (True Counts):
1. Prof-specialty             - True:  4140, Noisy:  4071.04
2. Craft-repair               - True:  4099, Noisy:  4069.73
3. Exec-managerial            - True:  4066, Noisy:  4099.41
4. Adm-clerical               - True:  3770, Noisy:  3769.40
5. Sales                      - True:  3650, Noisy:  3669.43
```

(a) What is the sensitivity of your scoring function?

Answer: The sensitivity is 1.

Why?

- Our function simply counts how many people represent each job.
- Imagine we add one person to the list.
- This person adds 1 to the count of their specific job.
- The counts for all other jobs stay the same (change is 0).
- So, the maximum change for any count is 1.

(b) What is the total privacy cost for most_common_occupation and why?

Answer: The total privacy cost is ε (epsilon).

Why?

- We use a rule called Parallel Composition.
- Think of each occupation as a separate box.

- One person fits into exactly one box (because nobody has two jobs in this data).
- Since the boxes are separate (disjoint), the privacy cost does not add up.
- We can use the full budget **ε** for each box without splitting it.
- Therefore, the total cost stays **ε.**

Task 4: Differentially private sums (4 points)

Output:

```
TASK 4: Differentially Private Sum (Capital Gain)
Calculating sum with epsilon = 0.04...
RESULTS:
--------------------------------------------------
Clipping Bound used:      100000
Sensitivity:              100000
Privacy Budget (ε):       0.04

--- Sum Values ---
True Sum (Clipped):       35,089,324
DP Sum (Noisy):           33,662,198
Noise Added:              -1,427,126
Relative Error:           4.07%
```

(a) What clipping parameter did you use and why?

Answer: I used a clipping parameter of **100,000**.

Why?

- **Domain knowledge:** In the Adult dataset documentation, *Capital Gain* values range from 0 to 99,999.
- **No bias:** By clipping at 100,000, I avoid cutting off valid values, so the sum remains accurate and unbiased.
- **Valid DP bound:** This gives us a clean, safe upper bound on each individual's possible contribution without inspecting private data.

(b) What is the sensitivity of the query and how is it bounded?

Answer: The sensitivity is **100,000**.

Explanation:

- Sensitivity Δf is the maximum contribution a single person can make to the sum.
- Since all values are clipped at 100,000, one person can change the sum by **at most 100,000**.

- Therefore, clipping ensures the sensitivity is strictly bounded by this value.

(c) Argue that your definition has a total privacy cost of ε.

Answer: The total privacy cost is exactly **ε**.

Why?

- The function makes **one query** (the sum of clipped Capital Gain).
- It applies the **standard Laplace mechanism** with scale

$$b = \frac{\Delta f}{\epsilon} = \frac{100000}{\epsilon}.$$

- The Laplace mechanism is known to provide **ε-differential privacy** for any numeric query with sensitivity Δf.
- Since there is no composition of multiple DP queries, there is no additional privacy cost.

Thus, the mechanism satisfies **ε-DP**.

Task 5: Sensitivity

Output:

```
TASK 5: Differencing Attack with Differential Privacy
Running protected differencing attack with epsilon = 1.0...

--- Attack Results ---
Target Person's True Age:      39 years
Attack Result (DP):            -295.64 years
Noise Added (Protection):      -334.64 years

--- Privacy Parameters ---
Total Privacy Budget (ε):      1.0
Budget per Query (ε/2):        0.5
Sensitivity:                   125

--- Attack Evaluation ---
✓ Attack PREVENTED: Noise (334.64 years) protects true age

--- Multiple Runs (showing noise randomness) ---
Run 1: Attack result = 44.93 years, Error = 5.93 years
Run 2: Attack result = -268.49 years, Error = 307.49 years
Run 3: Attack result = 664.96 years, Error = 625.96 years
```

Explain: What is the sensitivity of the differencing_attack query defined above, and why?

Answer: The sensitivity is the **maximum possible age** (for example, 125).

Why?

- **The Math:** The attack calculates Total Sum of Ages minus Sum of Ages without Karrie. The result of this calculation is exactly **Karrie's Age**.

- **The Definition:** Sensitivity is the maximum amount the result can change if one person's data changes.
- **The Reason:** If we change Karrie's age in the database (for example, from 0 to 125), the result of the attack also changes by that exact amount.
- **Conclusion:** Since a person's age can mathematically be any number up to a maximum limit (like 125), the sensitivity is equal to that **maximum limit**.